

Lab 3: Introduction to Vectors **Solutions**

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, September 16th, 2026

Name: _____

username: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

In each category, select the option that best describes your (honest) thoughts towards this class so far.

i. Difficulty

- ☐ Way harder than I expected
- ☐ Harder than I expected
- ☐ About as hard as I expected
- ☐ Easier than I expected
- ☐ Way easier than I expected

ii. Interest

- ☐ Way more interesting than I expected
- ☐ More interesting than I expected
- ☐ About as interesting as I expected
- ☐ Less interesting than I expected
- ☐ Way less interesting than I expected

Any immediate action items we can take to make your experience better?

Activity 1: Transformed Data

Let's begin by reviewing how transforming a dataset affects a fitted simple linear model.

Suppose we're given a dataset of n points, $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, where $\bar{x}$ is the mean of $x_1, x_2, \dots, x_n$ and $\bar{y}$ is the mean of $y_1, y_2, \dots, y_n$.

Using this dataset, we create a *transformed* dataset of n points, $(x'_1, y'_1), (x'_2, y'_2), \dots, (x'_n, y'_n)$, where:

$$x'_i = 4x_i - 3 \quad y'_i = y_i + 24$$

So the transformed dataset is of the form

$$(4x_1 - 3, y_1 + 24), (4x_2 - 3, y_2 + 24), \dots, (4x_n - 3, y_n + 24)$$

We decide to fit a simple linear model $h(x'_i) = w_0 + w_1 x'_i$ on the transformed dataset using squared loss. We find that $w_0^* = 7$ and $w_1^* = 2$, so $h^*(x'_i) = 7 + 2x'_i$.

- a) Suppose we were to fit a simple linear model through the original dataset, again using squared loss. What would the optimal slope on the original dataset be?

Solution: 8.

Relative to the dataset with x' , the dataset with x is compressed by a factor of 4, so the slope increases by a factor of 4: $2 \cdot 4 = 8$.

Concretely, this can be shown by looking at the formula for the new slope:

$$\begin{aligned} 2 &= r \frac{\sigma_{y'}}{\sigma_{x'}} \\ 2 &= r \frac{\sigma_y}{4\sigma_x} \\ 8 &= r \frac{\sigma_y}{\sigma_x} \end{aligned}$$

so the original slope is 8.

- b) Recall, the model $h^*(x'_i) = w_0 + w_1 x'_i$ was fit on the transformed dataset, introduced above. $h^*(x'_i)$ happens to pass through the point $(\bar{x}, \bar{y})$. What is the value of $\bar{x}$? Give your answer as an integer with no variables. *Hint: What else does $h^*(x'_i)$ pass through?*

Solution: $h^*(x'_i)$ is guaranteed to pass through $(\bar{x}', \bar{y}')$, where $\bar{x}'$ is the mean of the x' values and $\bar{y}'$ is the mean of the y' values.

Let's see what that looks like as an equation:

$$w_0^* + w_1^* \bar{x}' = h^*(\bar{x}')$$

$$7 + 2\bar{x}' = h^*(\bar{x}')$$

$$7 + 2\bar{x}' = \bar{y}'$$

Now write $\bar{x}'$ and $\bar{y}'$ in terms of $\bar{x}$ and $\bar{y}$:

$$\bar{x}' = 4\bar{x} - 3 \quad \bar{y}' = \bar{y} + 24$$

Substitute these into the equation above:

$$7 + 2(4\bar{x} - 3) = \bar{y} + 24$$

The problem also tells us that $h^*(x'_i)$ passes through $(\bar{x}, \bar{y})$, so

$$2\bar{x} + 7 = \bar{y}$$

Now subtract:

$$7 + 2(4\bar{x} - 3) - (2\bar{x} + 7) = \bar{y} + 24 - \bar{y}$$

$$7 + 8\bar{x} - 6 - 2\bar{x} - 7 = 24$$

$$6\bar{x} = 30$$

$$\bar{x} = 5$$

Recap: Vectors

Now that we've reviewed simple linear regression, let's turn to vectors. The next three activities explore vector lengths, linear combinations, and their implementation in NumPy.

- ([Chapters 3.1](#) and [3.2](#)) The **norm** of a vector $\vec{v} \in \mathbb{R}^n$ measures its length:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

This is the default norm for vectors in $\mathbb{R}^n$, but other norms exist.

- ([3.1](#)) A **linear combination** of the vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$ is any vector that can be written as

$$a_1\vec{v}_1 + a_2\vec{v}_2 + \cdots + a_d\vec{v}_d$$

where $a_1, a_2, \dots, a_d$ are scalars. We can think of this as taking bits of each vector and adding them together. The a_i 's are called the **coefficients** of the linear combination.

Activity 2: Triangle Inequality

We first use norms to compare the length of a sum with the lengths of the original vectors. The triangle inequality states that for any two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$:

$$\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$$

- a) For the vectors $\vec{u} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} -1 \\ -3 \end{bmatrix}$, verify that the triangle inequality holds. That is, show that the left-hand side is less than or equal to the right-hand side.

Solution:

First, let's find $\|\vec{u} + \vec{v}\|$.

$$\begin{aligned}\|\vec{u} + \vec{v}\| &= \left\| \begin{bmatrix} 4 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ -3 \end{bmatrix} \right\| \\ &= \left\| \begin{bmatrix} 4 - 1 \\ 3 - 3 \end{bmatrix} \right\| \\ &= \left\| \begin{bmatrix} 3 \\ 0 \end{bmatrix} \right\| \\ &= \sqrt{3^2 + 0^2} \\ &= \sqrt{9} \\ &= 3\end{aligned}$$

Also, $\|\vec{u}\| = \sqrt{4^2 + 3^2} = 5$. What's $\|\vec{v}\|$?

$$\begin{aligned}\|\vec{v}\| &= \left\| \begin{bmatrix} -1 \\ -3 \end{bmatrix} \right\| \\ &= \sqrt{(-1)^2 + (-3)^2} \\ &= \sqrt{1 + 9} \\ &= \sqrt{10}\end{aligned}$$

So, the triangle inequality claims that

$$\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$$

which, here, is

$$3 \leq 5 + \sqrt{10}$$

This is true, since 5 alone is greater than 3, so $5 + \sqrt{10}$ is surely also greater than (or equal to) 3.

- b) Find two **different** vectors $\vec{x}, \vec{y} \in \mathbb{R}^2$ such that the triangle inequality achieves **equality**, i.e. where

$$\|\vec{x} + \vec{y}\| = \|\vec{x}\| + \|\vec{y}\|$$

What is the relationship between the $\vec{x}$ and $\vec{y}$ you found?

Solution:

Example: let $\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$. Then,

$$\|\vec{x} + \vec{y}\| = \left\| \begin{bmatrix} 3 \\ 3 \end{bmatrix} \right\| = \sqrt{3^2 + 3^2} = 3\sqrt{2}$$

$$\|\vec{x}\| + \|\vec{y}\| = \sqrt{1^2 + 1^2} + \sqrt{2^2 + 2^2} = \sqrt{2} + \sqrt{8} = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$$

So, in this case, the triangle inequality achieves equality. What you'll notice is that $\vec{x}$ and $\vec{y}$ point in the same direction, i.e. $\vec{y} = 2\vec{x}$.

Activity 3: Linear Combinations

Next, we'll look at which vectors we can form by scaling and adding other vectors.

Let $\vec{u} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} -1 \\ -3 \end{bmatrix}$, and $\vec{w} = \begin{bmatrix} -6 \\ 9 \end{bmatrix}$.

- a) Find values of a and b such that $a\vec{u} + b\vec{v} = \vec{w}$. By finding a and b , you have written $\vec{w}$ as a **linear combination** of $\vec{u}$ and $\vec{v}$.

Solution:

We can pose this problem as solving a system of equations. By scalar multiplication, we have:

$$\begin{aligned} a \begin{bmatrix} 4 \\ 3 \end{bmatrix} + b \begin{bmatrix} -1 \\ -3 \end{bmatrix} &= \begin{bmatrix} -6 \\ 9 \end{bmatrix} \\ \begin{bmatrix} 4a \\ 3a \end{bmatrix} + \begin{bmatrix} -b \\ -3b \end{bmatrix} &= \begin{bmatrix} -6 \\ 9 \end{bmatrix} \\ \begin{bmatrix} 4a - b \\ 3a - 3b \end{bmatrix} &= \begin{bmatrix} -6 \\ 9 \end{bmatrix} \end{aligned}$$

The vector equation on the last line is equivalent to the system of equations:

$$\begin{cases} 4a - b = -6 \\ 3a - 3b = 9 \end{cases}$$

So, we just need to solve this system of equations to find a and b .

To do so, we can multiply the first equation by 3 to get:

$$\begin{cases} 12a - 3b = -18 \\ 3a - 3b = 9 \end{cases}$$

Then, we can subtract the second equation from the first to get:

$$9a = -27 \implies a = -3$$

Substituting $a = -3$ back into the second equation, we get:

$$3(-3) - 3b = 9 \implies -9 - 3b = 9 \implies -3b = 18 \implies b = -6$$

So, we have $a = -3$ and $b = -6$.

To verify that this solution works, we can substitute $a = -3$ and $b = -6$ back into the original equation:

$$(-3) \begin{bmatrix} 4 \\ 3 \end{bmatrix} + (-6) \begin{bmatrix} -1 \\ -3 \end{bmatrix} = \begin{bmatrix} -12 \\ -9 \end{bmatrix} + \begin{bmatrix} 6 \\ 18 \end{bmatrix} = \begin{bmatrix} -6 \\ 9 \end{bmatrix} = \vec{w}$$

- b) Now, try and write $\vec{w}$ as a linear combination of $\vec{u}$, $\vec{v}$, and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$. In other words, try and find values of a , b , and c such that

$$a \begin{bmatrix} 4 \\ 3 \end{bmatrix} + b \begin{bmatrix} -1 \\ -3 \end{bmatrix} + c \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \vec{w}$$

What happens? Why?

Solution:

We can start by trying to solve the corresponding system of equations:

$$\begin{cases} 4a - b + 2c = -6 \\ 3a - 3b + c = 9 \end{cases}$$

There are 2 equations and 3 unknowns, **which means there are infinitely many solutions for a , b , and c .**

What's the linear algebra reason for this?

- With just $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -3 \end{bmatrix}$, you can already create any other vector in $\mathbb{R}^2$. That is, any vector in $\mathbb{R}^2$ can be written as a linear combination of $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -3 \end{bmatrix}$.
- That is, for **any** vector $\vec{w} \in \mathbb{R}^2$ (not just the one in this question), there exist **unique values** of a and b such that

$$a \begin{bmatrix} 4 \\ 3 \end{bmatrix} + b \begin{bmatrix} -1 \\ -3 \end{bmatrix} = \vec{w}$$

- Since $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -3 \end{bmatrix}$ already can create any other vector in $\mathbb{R}^2$, adding $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ to the linear combination doesn't "unlock" any new vectors — we can still create any other vector in $\mathbb{R}^2$.
- But, because $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ already can be created using $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -3 \end{bmatrix}$, adding it to the linear combination makes it so that there are infinitely many solutions for a , b , and c in

$$a \begin{bmatrix} 4 \\ 3 \end{bmatrix} + b \begin{bmatrix} -1 \\ -3 \end{bmatrix} + c \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \vec{w}$$

If there are infinitely many solutions, how do we find them? Let's treat c as a free variable, and solve for a and b in terms of c .

$$\begin{cases} 4a - b + 2c = -6 \\ 3a - 3b + c = 9 \end{cases}$$

Multiplying the first equation by 3 gives us:

$$\begin{cases} 12a - 3b + 6c = -18 \\ 3a - 3b + c = 9 \end{cases}$$

Subtracting the second equation from the (new) first gives us:

$$9a + 5c = -27 \implies a = -3 - \frac{5}{9}c$$

Similarly, multiplying the first equation by 3 and the second equation by 4 gives us:

$$\begin{cases} 12a - 3b + 6c = -18 \\ 12a - 12b + 4c = 36 \end{cases}$$

Subtracting the (new) second equation from the (new) first gives us:

$$9b + 2c = -54 \implies b = -6 - \frac{2}{9}c$$

So, the values of a , b , and c that satisfy

$$a \begin{bmatrix} 4 \\ 3 \end{bmatrix} + b \begin{bmatrix} -1 \\ -3 \end{bmatrix} + c \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -6 \\ 9 \end{bmatrix}$$

are

$$a = -3 - \frac{5}{9}c, \quad b = -6 - \frac{2}{9}c, \quad c = c, c \in \mathbb{R}$$

c can be anything, which is why there are infinitely many solutions. If we let $c = 0$, then we get back $a = -3$ and $b = -6$ from part a). But, say, if we let $c = -9$, then we get $a = 2$ and $b = -4$, which also works:

$$2 \begin{bmatrix} 4 \\ 3 \end{bmatrix} - 4 \begin{bmatrix} -1 \\ -3 \end{bmatrix} + (-9) \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix} - \begin{bmatrix} -4 \\ -12 \end{bmatrix} + \begin{bmatrix} -18 \\ -9 \end{bmatrix} = \begin{bmatrix} -6 \\ 9 \end{bmatrix} = \vec{w}$$

c) Now, try and write $\vec{w}$ as a linear combination of $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -4 \\ -2 \end{bmatrix}$. What happens? Why?

Solution:

Note $\begin{bmatrix} -4 \\ -2 \end{bmatrix} = -2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, which means these vectors point in the same direction, or lie on the same line. (The formal term is that these vectors are **collinear**.)

Since $\vec{w} = \begin{bmatrix} -6 \\ 9 \end{bmatrix}$ is not a scalar multiple of $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ (ratios $-6/2 = -3$ vs. $9/1 = 9$ disagree), **no solution exists!**

To conclude, because the two vectors $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -4 \\ -2 \end{bmatrix}$ are collinear, it is impossible to write $\vec{w}$ as a linear combination of them. The only possible linear combinations are of the form $c \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ for some $c \in \mathbb{R}$.

Activity 4: Arrays in NumPy

Now, put vector operations into practice using NumPy arrays. Instead of writing code in a separate Jupyter Notebook for this lab, you will interact with the code cells that exist in the course notes.

In particular, go to [Chapter 3.2](#) of the course notes, scroll all the way to the bottom, and complete **Activity 5** there. To get checked off, show your lab TA that you've completed the activity — there's no need to submit your code anywhere.

Introduction: The Dot Product

So far, we've measured vector lengths and formed linear combinations. The dot product gives us a way to compare the directions of two vectors. We will study this in tomorrow's lecture; use the definitions below to explore it in the next two activities.

- (3.3) The **dot product** of two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ is defined as:

$$\vec{u} \cdot \vec{v} = \begin{bmatrix} u_1 \\ u_2 \\ \dots \\ u_n \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

The result is a **scalar**, not another vector.

- (3.3) The dot product also has a geometric definition, involving the norms (lengths) of two nonzero vectors and the angle between them:

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$$

- (3.3) The key takeaway from the dot product is that it tells us how similar the directions of two vectors are. When two vectors have a dot product of 0, they are **orthogonal**. For two nonzero vectors, this means that the angle between them is 90° .

Activity 5: The Dot Product

For each pair of vectors below, (1) draw them on the grid to the right and (2) compute their dot product using the component-wise definition.

a) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Solution:

$$8 \cdot 1 + 6 \cdot 0 = 8.$$

b) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 0 \end{bmatrix}$

Solution:

$$8 \cdot (-2) + 6 \cdot 0 = -16.$$

c) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 6 \\ 8 \end{bmatrix}$

Solution:

$$8 \cdot 6 + 6 \cdot 8 = 48 + 48 = 96.$$

d) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$

Solution:

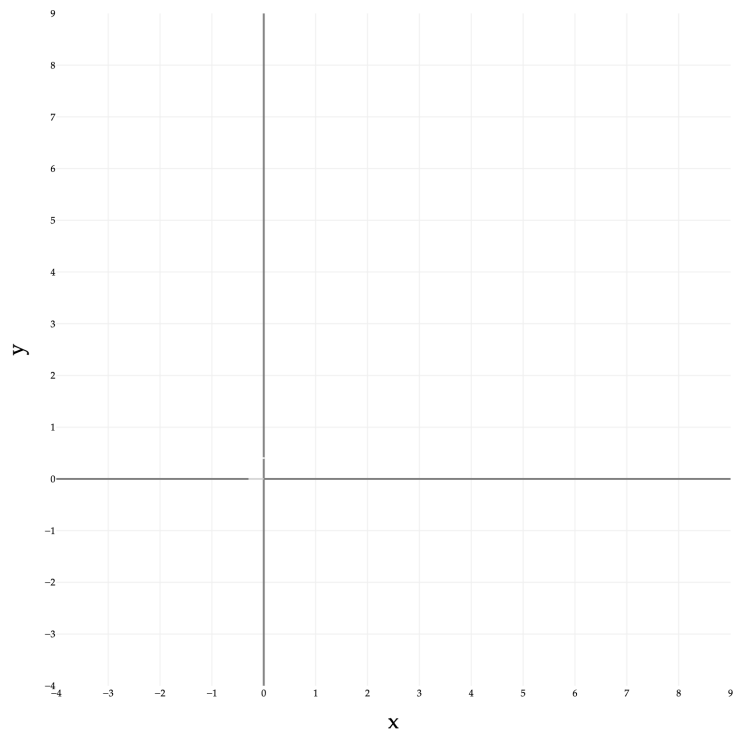
$$8 \cdot 8 + 6 \cdot 6 = 64 + 36 = 100.$$

e) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 4 \end{bmatrix}$

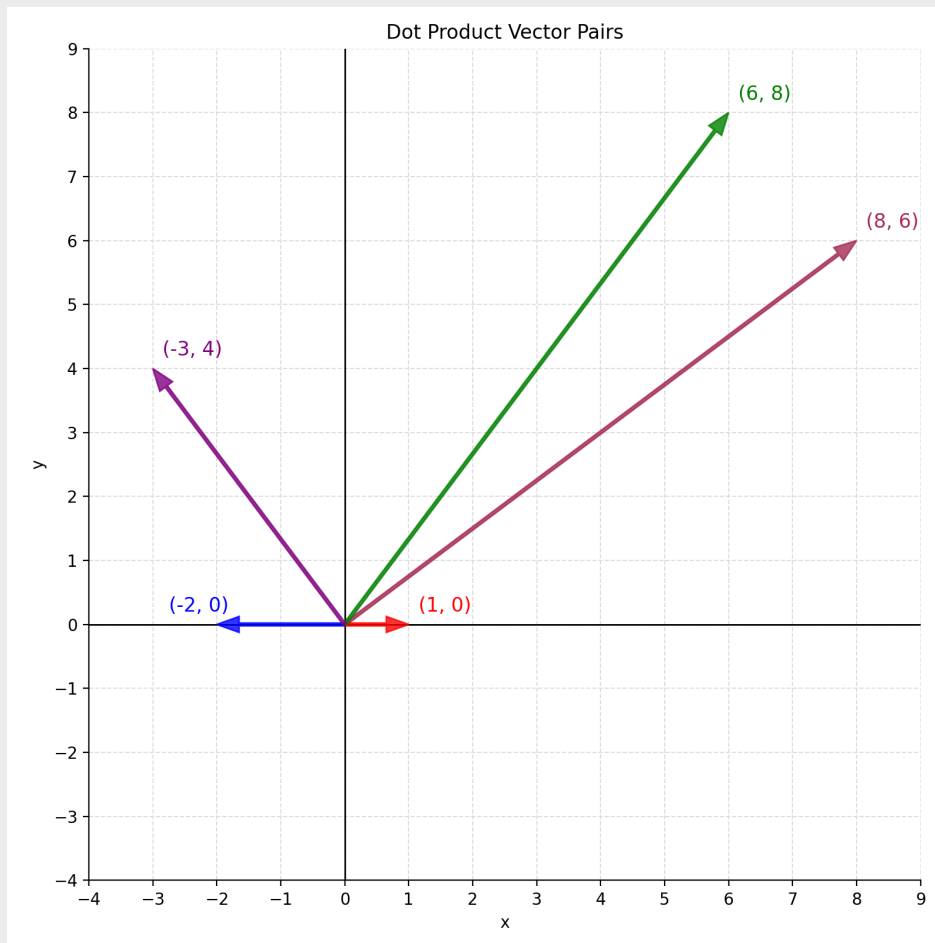
Solution:

$$8 \cdot (-3) + 6 \cdot 4 = -24 + 24 = 0.$$

Since the dot product is 0, the vectors are **orthogonal**.



Solution:



Activity 6: Angles and Orthogonality

In the previous activity, we computed dot products using vector components. Now, we will connect those calculations to angles using the geometric definition of the dot product.

$$\vec{u} \cdot \vec{v} = \begin{bmatrix} u_1 \\ u_2 \\ \dots \\ u_n \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$$

Let $\vec{w} = \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \end{bmatrix}$ and $\vec{x} = \begin{bmatrix} 9 \\ 1 \\ 2 \\ 3 \end{bmatrix}$.

- a) Find $\vec{w} \cdot \vec{x}$, $\|\vec{w}\|$, and $\|\vec{x}\|$.

Solution:

$$\vec{w} \cdot \vec{x} = 5 \cdot 9 + 0 \cdot 1 + (-4) \cdot 2 + 1 \cdot 3 = 45 + 0 - 8 + 3 = \boxed{40}.$$

$$\|\vec{w}\| = \sqrt{5^2 + 0^2 + (-4)^2 + 1^2} = \sqrt{25 + 0 + 16 + 1} = \boxed{\sqrt{42}}.$$

$$\|\vec{x}\| = \sqrt{9^2 + 1^2 + 2^2 + 3^2} = \sqrt{81 + 1 + 4 + 9} = \boxed{\sqrt{95}}.$$

- b) Using the results of part a), find the angle between $\vec{w}$ and $\vec{x}$. Leave your answer in the form $\cos^{-1}(\cdot)$.

Solution:

Using $\vec{w} \cdot \vec{x} = \|\vec{w}\| \|\vec{x}\| \cos \theta$,

$$\cos \theta = \frac{\vec{w} \cdot \vec{x}}{\|\vec{w}\| \|\vec{x}\|} = \frac{40}{\sqrt{42} \sqrt{95}}.$$

Therefore

$$\theta = \cos^{-1}\left(\frac{40}{\sqrt{42} \cdot \sqrt{95}}\right).$$

- c) What is $\cos(90^\circ)$? What does this have to do with orthogonality?

Solution:

$$\cos(90^\circ) = 0.$$

If the angle θ between $\vec{u}$ and $\vec{v}$ is 90° , then

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta = \|\vec{u}\| \|\vec{v}\| \cdot 0 = 0,$$

so the vectors are **orthogonal**. Conversely, if $\vec{u} \cdot \vec{v} = 0$ (and neither vector is the zero vector), then $\cos \theta = 0$ and $\theta = 90^\circ$.